Software Closeout Review Checklist

8 Digit Case #: _____ Plan Name: _____

DOPT: _____ Valuation Data Environment*: _____

Program ID #: _____ If there are multiple Program IDs, a separate Checklist is required for each Program being published.

When there are multiple checklists, each checklist should have the file name *SCR_CKLST - ####.pdf* where #### is the Program ID #.

BSRS Data Environment (if applicable)*: _____

Actuary: _____ Contract Actuary Name and Firm: _____

Actuarial Supervisor: _____ Statement Authoring Owner: _____

* If a different Valuation Data Environment and BSRS Data Environment are specified above, the Valuation Data Environment will be used for preserving the valuation values in the Final Valuation Dataset. The BSRS Data Environment will be used for CCS statement generation, for the production BSRS program, and for Data Validation. However, if there are multiple data environments for this case that are to be used in a different manner than previously described, please specify in the General Comments box on Page 10 and check the following box. ☐

☐ The Valuation has been reviewed & approved by ASD management. The Valuation Data load is irreversible!

☐ Participant counts & PVs agree with Data Validation Summary Report, Actuarial Case Memo, and CMS. Explain differences (e.g., different treatment of non-members, Data Validation limitations for Deferred Beneficiaries, APs, etc.) in the ***Appendix*** located at the end of this document.

☐ The CMS Final Participant Liabilities milestone has been finalized. If not, an email from the processing management has been sent to TRMD that includes the liability totals to be used for verification.

☐ All present value fields (i.e., PVMB_TERM, LS_TERM, etc.) should be integer values within the database.

Document	Participant Count (excluding non-members)	Termination PV Total	Lump Sum Discriminator Total	Present Value EECs
CMS				N/A
Actuarial Case Memo				N/A
Data Validation Summary Report				

DATA VALIDATION WARNINGS & ERRORS

☐ Explanation for all errors that cannot be fixed and warnings are explained in the ***Appendix*** located at the end of this document.

☐ Save Data Validation Summary Report and the Data Validation Detail Report as PDF's to ASTD > TRMD Services Program Site > Service Requests >Case Number >Final Valuation.

Case #: Plan Name:

VALUATION DATABASE

☐ The lock on the Valuation Data Environment (and the BSRS Data Environment if applicable) has been released using the "Release Save to Data Environment Hold" option under Session Management in the Calculation Authoring section of the BCV Web Portal.

☐ When possible, pre-defined fields were used instead of user-defined fields.

☐ Each populated user-defined field has been described in the UD table.

☐ Non-spouse beneficiary data and alternate payee data has been stored in the designated fields and not in the spouse fields. [3]

☐ If the QPSA benefit is non-level, the benefit in XRD_MB_QPSA_TERM should be the amount payable during the last payment stream. If the participant has multiple benefits that are administered separately (i.e., multiple XRD benefit fields), these QPSA fields are populated with the values based on the XRD benefit with the largest DOPT value. [2]

☐ The fields XRD_MB_TERM, FORM_CODE, SPC_VAL, and MTHS have been populated with the appropriate values for the expected non-disability benefit payable at the date populated in the field AUX_DIS_CHANGE_DATE in any records that are for participants still receiving an auxiliary disability benefit at the time of the Data Validation Load (the fields AUX_DIS_BEN and AUX_DIS_CHANGE_DATE must be populated in any records that are for participants receiving auxiliary disability benefits after DOPT). Note the expected QPSA benefit and form of annuity payable at the date populated in the field AUX_DIS_CHANGE_DATE must also be populated per the previous item. The fields DOR, FORM_CODE_ARD, SPC_ARD, MTHS_ARD are NULL if the participant is still receiving the auxiliary disability benefit at the time of the Data Validation Load, otherwise these fields are populated with the values corresponding to the non-disability benefit commenced after DOPT. [2]

☐ If the benefit has not yet commenced at the time of the Data Validation Load for records classified as in pay at DOPT (RETSTAT = '1') because the required begin date is before DOPT, the fields DOR, FORM_CODE_ARD, SPC_ARD, and MTHS_ARD are NULL and the field TERM_MB_ERD_NSF is populated with the benefit payable at the pre-DOPT required benefit date and the field BDERDLEV is populated with the pre-DOPT required begin date. [1]

☐ The field MANDCWI is populated with the appropriate balance as of DOPT for any record in which a post-retirement death benefit is potentially applicable as a result of a refund feature present with the form of annuity (including those in pay and future payees). This field must be populated with the balance at DOPT (not the balance at benefit commencement date) for any records in pay prior to DOPT if there is still a positive balance remaining at DOPT. [2]

☐ Valuation database has been updated with Post-DOPT information (e.g., spouse information, DOR, etc.) on . [1, 3] Update should be within last 30 days.

☐ BR_IND is set to N for all records where no future recalculation is applicable (e.g., benefit is already in pay). [3]

☐ BS_IND is set to N for all records where no statement is generated by BSRS (manual statements or situations where a statement is not required, e.g., retirees with no benefit change). [3]

☐ All active groups have a valid SQL definition. Also, all SQL definitions do not include any actual PII (customer names and SSNs) and do not reference any BCV database fields containing PII (SSN, FNAME, LNAME. SSSN, SFNAME, SLNAME, etc.). [3]

☐ All input data changes made during the valuation have been updated in the Database Building module.

☐ IPVFB supplemental data for this case has been copied to the staging area through TRMD Services. The IPV1500 and IPV6100 reports have been reviewed, and the Supplemental Data Extract has been confirmed by the actuary.

1 Requirement for generating BD's via the Customer Communication System (CCS).
2 Requirement for determining accurate future liability via the IPVFB system.
3 Requirement for BSRS program.

Case #: _____ Plan Name: _____

VALUATION SPREADSHEET

- ☐ None of the cells with formulas are formatted as text.
- ☐ All range names have valid references and are applicable for the case.
- ☐ All participants' data has been removed from the V1.xlsx spreadsheet.
- ☐ Fixed invalid links to ATPBGC Excel Addin.
- ☐ Cells or ranges containing asset allocation ratios, load percentages, etc. which are referenced by formulas elsewhere in the spreadsheet should contain only values (i.e., no formulas).
- ☐ Excel protection turned off for V1 spreadsheet.
- ☐ I/O/B row has been set to 'I' for CALC_INDICATOR on the QPSA row when the BSRS_VERSION=4.
- ☐ I/O/B row has been populated with I, O, B, N or P for all fields.
- ☐ I/O/B row has been set to 'I' for MSTAT and QPSA_INDICATOR in V1 sheets used for recalculation. [3]
- ☐ When the V1 spreadsheet does not include QPSA calculation rows, the I/O/B row has been set to 'B' for the fields XRD, FORM_CODE, SPC_VAL, MTHS, LS_QPSA, and LS_TERM in the XRD calculation rows. The I/O/B row can be set to 'B' for other fields but caution should be used since such fields will not recalculate based on the data entered on the ARD data entry screen. When the V1 spreadsheet includes QPSA calculation rows, LS_QPSA has been set to an 'I' field in the XRD calculation rows (if it is present), and set to an 'O' field in the QPSA calculation rows. Note, when the V1 spreadsheet includes the QPSA calculation rows, XRD_MB_QPSA_TERM is generally also set to an 'I' field in the XRD calculation rows (if it is present), and set to an 'O' field in the QPSA calculation rows.
- ☐ The spreadsheet does not contain any imbedded VisualBASIC. [3]
- ☐ No <u>actual</u> PII (customer names and SSNs) has been included on any active or inactive Excel sheets or in any Excel formulas. Additionally, no Excel formulas reference any BCV database fields containing PII (SSN, FNAME, LNAME, SSSN, SFNAME, SLNAME, etc.).
- ☐ VAL_SDOB is programmed for spouse and non-spouse beneficiaries. [3]
- ☐ Used latest versions of functions: XRD2, XRA2, PBGCBFCF2, PBGCJSF , NPVF2, LLF, SXRA2, and MAXLIM.
- ☐ No Excel sheets in the V1.xlsx file contain circular formula references. "Circular References" should be grayed out on the "Error Checking" drop-down menu on the Formula tab in Excel.
- ☐ Removed any extraneous Calculation Authoring groups or sheets from V1 spreadsheet (e.g., if no deferred beneficiaries in case, no sheet or Calculation Authoring group included for deferred beneficiaries).
- ☐ Multiple Calculation Authoring groups with identical programming are combined into a single Calculation Authoring group whenever possible.
- ☐ The key fields C1.CASE, and C1.BCV_REC_ID are included as input fields for each calculation (e.g., XRD, NRD, ERD, etc.) on each Calculation Authoring group sheet (e.g., Retirees, Beneficiaries, Deferreds, etc.). [3]
- ☐ Late retirement factors and Benefit Form Conversion Factors are calculated in the V1 using a formula instead of a table of factors when possible. If a table of factors is necessary, the table is values only except for the number of cells (usually one) necessary to provide the formula used to calculate the factors.
- ☐ A separate version of the V1 file was used to calculate valuation liabilities <u>only</u> (not for the BSRS program) and is being provided for archiving purposes with the file name ########V2.XLSX where ######## is the eight-digit case number.

1 Requirement for generating BD's via the Customer Communication System (CCS).
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BSRS

☐ All records with BR_IND = Y or (BS_IND=Y and DOR is not NULL) satisfy the criteria for only one group immediately upon selection.

☐ The BSRS Statement and OFA layouts do not contain any SSN or names.

☐ All QPSA recalculation statements include information about the QPSA lump sum in the benefit summary section. When displayed on QPSA recalculation statements, the calculation of LS_QPSA has been tested to ensure that it is based on the actual spouse DOB and on the plan's defined QPSA form (which may be different than the normal married form).

☐ Optional Forms of Annuity output has been enabled for all participants with exceptions noted in the Actuarial Case Memo (R1.doc).

☐ Is the OFA_CONFIG_TYPE variable set to STANDARD CONFIGURATION in the Programs Variable section? If no, explain the reason for the Custom OFA Configuration in the General Comments Box of the ***Appendix.***

☐ Is the LATEST ALLOWABLE RETIREMENT DATE variable enabled in the Program Variable section? If yes, explain the reason in the General Comments Box of the ***Appendix***.

☐ BSRS handles spouse and non-spouse beneficiaries (non-spouse beneficiary data in designated fields).

☐ An error message is provided on the recalculation statement if the user is running a QPSA calculation for the surviving spouse of a participant alive at DOPT when an estimated lump sum has been paid and the residual lump sum is less than or equal to $5,000 ($7,000 if DOPT is on or after 01/01/2024).

☐ No fields are included on the base data entry screen that are not relevant to the benefit calculation (i.e., if a change to the value in a field does not have the possibility of impacting the benefit calculation, the field should not be included on the base data entry screen). Note, fields that contain values displayed on the output but that do not affect the benefit calculation can be included on the base data entry screen (e.g., FNAME, LNAME, SSN, etc.)

☐ Appropriate limits have been set for all fields included on the base data entry screen to ensure that values entered will not lead to scenarios that are outside the scope of the programming.

☐ If there is a finite set of valid entries for a text field included on the base data entry screen (e.g. PSEX, PHS, PRSAC, etc.), the field has been setup with a drop-down menu containing the valid entries that the user can select for the field (e.g. for PSEX the valid entries would be M or F).

☐ The base data entry screen includes the fields PCI and PCCI with the Description "Participant's Customer ID:" and "Participant's Case Customer ID:", respectively.

☐ There is BSRS guidance in section 25 of the actuarial case memo that provides additional details for the fields included on the base data entry screen or for other issues related to benefit administration.

EMPLOYEE CONTRIBUTIONS (BSRS)

☐ Are employee contributions applicable (participants with EEC at DOPT and plan allows withdrawal)?

☐ The recalculation statement and optional forms output show the net benefit alone if the contributions have been withdrawn, the gross benefit alone if the contributions cannot be withdrawn without taking a full lump sum, or both the gross and net benefit if the contributions have not yet been withdrawn.

☐ The recalculation statement shows the residual lump sum if it is less than or equal to $5,000 ($7,000 if DOPT is on or after 01/01/2024), LS_TERM is greater than $5,000 ($7,000 if DOPT is on or after 01/01/2024), and the employee contributions have been withdrawn.

☐ An error message is provided on the recalculation statement if the user indicates that employee contributions have been withdrawn when LS_TERM is less than or equal to $5,000 ($7,000 if DOPT is on or after 01/01/2024).

1 Requirement for generating BD's via the Customer Communication System (CCS).
2 Requirement for determining accurate future liability via the IPVFB system.
3 Requirement for BSRS program.

Case #: Plan Name:

EMPLOYEE CONTRIBUTIONS (BSRS) CONT'D

☐ An error message is provided on the recalculation statement if the user is running a QPSA calculation for the surviving spouse of a participant alive at DOPT when the contributions have been withdrawn and the residual lump sum is less than or equal to $5,000 ($7,000 if DOPT is on or after 01/01/2024).

☐ Base data screen includes fields that indicate whether EECs have been or will be withdrawn.

CCS LOAD

☐ If there are multiple Programs being published for this case, are there participants receiving a benefit statement from more than one Program?

☐ The BD benefit statements/worksheets for all records in which BS_IND = 'Y' are to be loaded to CCS (only select NA if no records have BS_IND = 'Y'). If NO is selected, specify the BD benefit statement group number created in Calculation Authoring that should be used to generate the BD benefit statements/worksheets to be loaded to CCS.

The load to CCS includes (only select NA if no records have BS_IND = 'Y'):

MANUAL STATEMENTS/WORKSHEETS (CCS LOAD)

☐ Are there manual statements/worksheets that are to be loaded to CCS (if YES is selected, the file ManualStmts.zip must be included in files provided).

☐ The filename for each manual BD benefit statement/worksheet is #######_BS.PDF or #######_WS.PDF where ####### is the seven-digit Case_Cust_ID value from Genesis for the recipient of the BD benefit statement/worksheet.

☐ If a manual BD statement/worksheet is provided for a record, "BS_IND = 'N' in that record.

ONLINE ESTIMATE

☐ Does this BSRS contain customers who are eligible for online estimates via MyPBA? If yes, the population of customers eligible for online estimates has been identified under the Online Estimate in the BSRS/BET Program Management section of the BCV Web Portal. For details on which customers are not eligible for online estimates, please visit the BCV Manual.

TESTING

BD Benefit Statements

☐ Run all statements to File to ensure that benefit statement generation does not fail for participants with BS_IND='Y'.

☐ Generate the BD benefit statement for each of the records listed in the BSRS Testing table below and save the BD benefit statement generated as a PDF file using the file name LNAME_FNAME_PP_#######.pdf.

Recalculation Output

☐ Run the recalculation tests listed below for each type of future payee that has BR_IND='Y' (note these tests can be run via Individual Recalculation or via Bulk Recalculation). Note that not all tests below are applicable to all test records selected by the actuary.

☐ Save recalculation outputs (recalculation statements and optional forms of annuity outputs) as PDF files using BCV standard file naming conventions. Some file naming examples are given following each test listed below.

BSRS_TEST ZIP File

☐ **Zip** the BD benefit statement and recalculation output PDF files generated into ***BSRS_Test.zip***. **Submit BSRS_*Test.zip*** with the files being archived for this case. The BD benefit statement and recalculation output for at least one record must be included in the ***BSRS_Test.zip*** submitted for the case.

Case #: Plan Name:

TESTING CONT'D

1. **XRD Test**: Run a recalculation at XRD to ensure that the benefit matches XRD_MB_TERM in the DOPT listings. The test should be run using the same marital status and beneficiary data used to calculate the XRD benefit for the valuation. The benefit on the recalculation statement showing the benefit in the valuation form of annuity should match the value in XRD_MB_TERM. Example Output Name:

   LNAME_FNAME_PP_########_XRD_BSNMFSP.pdf, LNAME_FNAME_PP_#######_XRD_OFSP.pdf

2. **ERD Test**: Run a recalculation at ERD to ensure that the benefit matches TERM_MB_ERD_NSF (benefit amount at the PBGC earliest benefit commencement date generally shown on the BD Statement). Verify that the benefit on the recalculation statement matches the benefit at the PBGC earliest benefit commencement date on the BD statement generated as part of the benefit statement testing from above. Note, ERD is often equal to the earliest retirement date under the plan to which the participant is entitled but can be later for participant's whose earliest retirement date under the plan is earlier than the BD issuance date. If the test is run for a married participant or for an unmarried participant with a non-spouse beneficiary, the benefit on the recalculation statement showing the benefit in the normal unmarried form of annuity should match the value in TERM_MB_ERD_NSF and the benefit at the PBGC earliest benefit commencement date on the BD statement. Example Output Name:

   LNAME_FNAME_PP_#######_ERD_BSNSF.pdf, LNAME_FNAME_PP_#######_ERD_OF.pdf

3. **EURD Test**: Run a recalculation at EURD to ensure that the benefit matches TERM_MB_EURD_NSF (amount generally pulled in the BD). Verify that the benefit on the recalculation statement matches the benefit on the BD statement generated as part of the benefit statement testing from above. Note, EURD is often equal to Normal Retirement Date (NRD) but can be earlier in plans that offer early unreduced retirement or can be later for participants whose NRD is earlier than the BD issuance date. If the test is run for a married participant or for an unmarried participant with a non-spouse beneficiary, the benefit on the recalculation statement showing the benefit in the normal unmarried form of annuity should match the value in TERM_MB_EURD_NSF and the benefit on the BD statement. Example Output Name:

   LNAME_FNAME_PP_#######_EURD_BSNSF.pdf, LNAME_FNAME_PP_#######_EURD_OF.pdf

4. **RBD Test**: Run a recalculation at the RBD (required begin date) to ensure that calculations at late retirements are working properly. The test can be executed for a married or unmarried participant with or without a non-spouse beneficiary. Example Output Name:

   LNAME_FNAME_PP_#######_RBD_BSNMFSP.pdf, LNAME_FNAME_PP_#######_RBD_OFSP.pdf

5. **1YR Test**: Run a recalculation with a one year old non-spouse beneficiary to ensure that calculations with beneficiaries of any age are working properly. Ensure that the non-spouse beneficiary is born after DOPT. The test can be executed for a married or unmarried participant. Example Output Name:

   LNAME_FNAME_PP_#######_1Yr_BSNMFBENF.pdf, LNAME_FNAME_PP_#######_1Yr_OFBENF.pdf

6. **QPSA Test**: Run a recalculation with a pre-retirement death to ensure that the calculations of QPSA benefits are working properly and that the lump sum language is displayed in the benefit summary section. Example Output Name:

   SLNAME_SFNAME_SP_#######_QPSA_BSQPSA.pdf, SLNAME_SFNAME_SP_#######_QPSA_OF.pdf.pdf

7. **EECIN, EECRET Test:** Run a recalculation with EEC left in the plan and run a recalculation with EEC returned to participant to ensure that calculations with and without EECs are working properly. The test can be executed for a married or unmarried participant with or without a non-spouse beneficiary. Example Output Name:

   LNAME_FNAME_PP_#######_EECRET_BSNMFSP.pdf, LNAME_FNAME_PP_#######_ EECRET_OFSP.pdf

Case #: _____ Plan Name: _____

Determination of test records is left to the judgment of the actuary. However, at a minimum a record from each Calculation Authoring group should be tested unless the programming for multiple groups is identical (best practice is to combine multiple Calculation Authoring groups into a single Calculation Authoring group whenever there is identical programming). Briefly define test records below:

Name	Description

Click here to download additional page for testing.

You must enter each test record identified above in the chart below. Check the tests executed for each test record. Any test that is applicable for the test record should be executed with the output saved as PDF following the naming conventions shown above. There is no need to enter benefit amounts on the checklist.

BSRS TESTING								
							Employee Contributions	
Name	XRD Benefit	ERD Benefit	EURD Benefit	RBD Benefit	1-Year Old Beneficiary	QPSA Benefit	EEC Withdrawn	EEC Left In

Case #: _____ Plan Name: _____

ARCHIVING

The following are Standard file names, and will be archived on the first level. Copy files to be archived to ASTD site in Sharepoint (ASTD > TRMD Services Program Site > Service Requests > > Final Valuation).
Click here to see complete list

If you ever need to modify these files, contact TRMD Services to update the files.

☐	R0.docx	Table of Contents for Actuarial Case Report Actuarial
☐	R1.docx	Case Report
☐	R5.docx	Plan Abstract
☐	PF.xlsx	Plan Factor Tables (one per sheet)
☐	V1.xlsx	Valuation Spreadsheet
☐	V2.xlsx	Copy of V1 File for Valuation Purposes Only
☐	LL.xlsx	DOPT Listing (listings in one Excel file)
☐	_ManualStmts.zip	Manual Statements (statements in PDF format)
☐	BSRS_Test.zip	Recalculation Output for Future Payees
☐	DataValidationSummary.pdf	BSRS Data Validation Summary
☐	DataValidationDetail.pdf	Data Validation Detail Report
☐	SCR_CKLST.pdf	Software Closeout Review Checklist

☐ All Word documents have been saved with all changes having been accepted (for scenarios where track changes had been enabled on the document).

☐ All Software files used for benefit calculations outside of the V1.xlsx file, including calculations used for producing manual statements, have been provided for archiving as non-standard files and are documented in NON-STANDARD FILES FOR ARCHIVE in the ***Appendix***. Archive must contain all files needed to reproduce all of the calculations for every termination benefit in the plan. These files must be copied to SharePoint (ASTD > TRMD Services Program Site > Service Requests > > Final Valuation). If you ever need to update these files, contact TRMD Services.

FILE CLEANUP

All incoming and outgoing files placed on the TRMD Services program site are subject to deletion by TRMD.

GS-13 or Higher Actuary Review and Approval

☐ *Retirement eligibility, benefits, lumps sums, etc. calculated by the BSRS program, have been reviewed and approved by a GS-13 or higher actuary.*

By checking the boxes on this checklist, the Actuary assigned to the case is certifying that the software complies with the standards detailed above and the data which is loaded to Genesis and the IPVFB Supplemental Table has been properly configured and is accurate. In electronically signing the box below, the GS-13 or higher actuary is approving the boxes selected in the checklist by the Actuary (e.g., whether or not Leo was used to track data changes, whether or not there are participants retired prior to the Data Validation load with J&S Pop-up annuity, etc.)

Note: Click on the validation button to confirm all required fields are filled and enable the signature field. **Validation**

GS-13 or Higher Actuary Reviewer Signature:

Click here for assistance with Digital signature

APPENDIX : EXPLANATIONS

Explanation for PV and participant count discrepancies

Explanation for Data Validation Warnings and Errors (include Ref #)

Non-Standard Files for Archive

General Comments